Hall-Effect Sensor Fault Diagnosis, Identification and Remedial Strategy in Permanent Magnet Brushless DC Drive

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***Abstract*—In this paper, a simple fault tolerant control for failure of one Hall Effect position sensor of PMBLDC drive is proposed in a closed loop control scheme to increase the reliability of drive. The proposed system is capable to detect and identify the breakdown of one Hall sensor and replaces the faulty sensor signal by a fictitious Hall signal which is generated by the microcontroller. Effectiveness of the remedial strategy is also proven by the correct performance of PMBLDC drive under faulty condition during various transition conditions. After occurrence of the fault, PMBLDC drive is unable to self-start with only two hall sensors. Therefore, a new starting methodology is also presented. The proposed control and starting method have been experimentally verified.**

***Index Terms*—Permanent Magnet Brushless DC (PMBLDC) Motor , Hall Sensor, Fault Diagnosis, Fault Tolerant Control**

I. Introduction

The popularity of PMBLDC motors has increased hugely in the last decade. They are being used in various applications ranging from industrial to home appliances because of the various advantages they offer like high efficiency, high power density, low noise, low maintenance cost, and simple control. The important constructional feature is that these motors have a permanent magnet rotor. In place of a mechanical commutator, these motors work on electronic commutation. For proper operation of these motors, information about rotor position is necessary. This can be obtained by the use of position sensors which can be Hall Effect sensors or optical ones. Encoder or various sensorless techniques elaborated in [1], for example: techniques based on back-emf sensing, back-emf integration, flux linkage estimation, freewheeling diode current detection etc. can also be used.

Since last few years, research has been going on for improving the sensorless algorithms. But, most of the sensorless algorithms do not work properly at low speeds and need open loop commutation for starting the motor. Some require the building of a virtual neutral point and some do not work for non-salient machines. Therefore, usage of Hall Effect sensors for rotor position detection is still prevalent for many applications.

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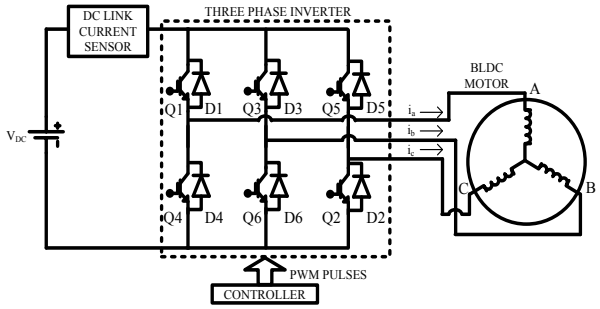
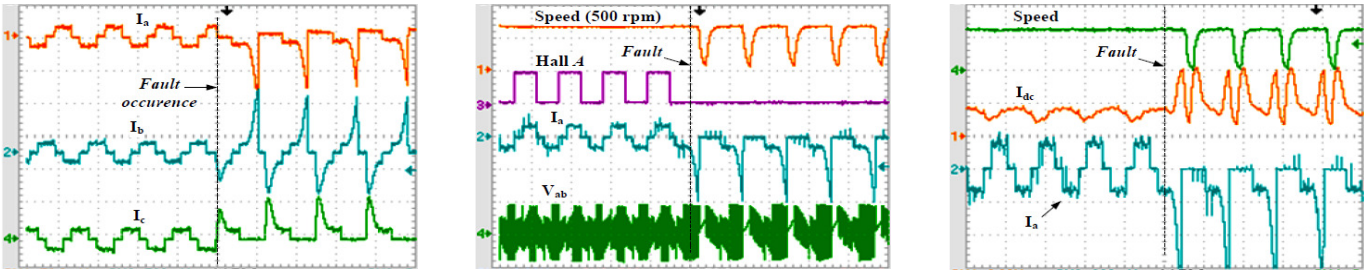


Fig. 1: Simplified BLDC Drive Scheme.

A typical PMBLDC motor requires three Hall signals to detect the rotor position for operation. The Hall sensors are ideally placed 120 electrical degrees apart. Electronic commutation of PMBLDC motor depends on the accurate detection of permanent magnet rotor position Fig. 1. Failure of Hall sensors directly affect the voltages applied to the PMBLDC motor and degrade the performance of overall motor drive by deteriorating current and torque ripple. Therefore, development of fault tolerant control system is necessary to increase the reliability of electric motor drive. Failure of Hall sensors may happen due to different reasons like aggressive environment, vibrations, circuit faults, etc [2]. It results in unavailability of the corresponding signals, thereby affecting the operation of the motor.

Many Hall sensor fault diagnosis and fault mitigation techniques have been proposed in literature [3] - [8]. In [3], many types of Hall sensor faults like single and double Hall sensor faults have been described. Different fault tolerant control schemes have been discussed. In [4], binary Hall effect sensor faults are investigated in PMBLDC drives and an effective methodology for their compensation is explored. In [5], an improved fault tolerant control (FTC) scheme based on fault diagnostic process (FDP) and vector-tracking observer is developed. A fault compensation method based on vector-tracking observer for mitigating single and multiple Hall-effect sensor faults is also discussed in [6]. In [7], wavelet signal



(a) (b) (c)

Fig. 2: Behavior of PMBLDC motor during one hall sensor fault ($H_A = 0$). Time Scale : 50 msec/div, Scale : a) Motor phase currents : I_a,I_b,I_c = 5 A/div b) Speed = 400 rpm/div, Hall Sensor Signal,H_A = 5 A/div, Phase current,I_a = 5 amps/div, Line Voltage,V_{ab} = 100 volts/div. c) Speed = 400 rpm/div, DC Link current, I_{dc} = 5 amps/div, Phase current,I_a = 5 amps/div.

theory is used as a diagnostic method, to identify either a faulty or a misplaced position sensor. In [8], the diagnostic method includes finding α-β plane hodographs of phase currents' vector and rotor position sensors' vector. After the detection and identification of a faulty sensor, next step is hardware or software reconfiguration to have fault tolerant operation of the drive. A summary of many of the compensation methods is given in [9].

This paper deals with the operation and behavior of PMBLDC drive in the event of failure of one Hall sensor. Fault diagnosis, identification and a remedial control strategy is proposed for smooth operation of the motor. A starting method for the motor with only two Hall sensors is elaborated. Transient response of the motor with the proposed control is also analyzed.

II. Behavior of Motor During the Failure of One Hall Effect Sensor

Breakdown of the Hall sensor causes its output signal to either remain constantly high (logic 1) or constantly low (logic 0). Therefore, the output signal does not change with the rotor position. The constant output causes change in the switching signals of VSI and thus, affecting the voltages applied for the excitation of the motor [10]. The affect of sensor faults on phase currents is discussed in [8].

Changes in the applied voltages cause changes in the stator phase currents. This in turn adversely affects the electrical torque of the motor as there is an increase in the torque ripple. Phase currents of PMBLDC motor running at 500 rpm under 1.2 Nm load torque are shown in Fig. 2(a) for a fault condition of $H_A = 0$. It is seen that motor starts drawing excessive currents in the event of the fault. Fig. 2(b) shows that after the fault occurrence, speed of the motor begins oscillating. The output of the faulty Hall sensor becomes constant at '0'. Even the line voltage, V_{ab} is getting distorted because the switching signals of VSI are affected due to the fault. It can be seen in Fig. 2(c) that DC Link current of the motor which is proportional to the developed torque also gets disturbed as its magnitude suddenly increases. From these speed and current characteristics, it is deduced that the working of the motor is disturbed because of the breakdown of position sensor.

III. Proposed Drive Strategy For Fault Analysis

The fault tolerant control system mainly should accomplish the tasks mentioned in Fig. 3. Firstly, the algorithm should detect the fault occurrence. Secondly, the faulty Hall sensor is identified. Finally, the mitigation and compensation techniques are implemented.

A. Fault detection

Each Hall senor signal of PMBLDC motor has a specific value at each instant of time with respect to rotor position. It can be seen through the switching sequence given in Table I that all three hall signals simultaneously be at either one or zero. For fault detection, a variable named H_{sum} is defined which is the sum of instantaneous value of all Hall signals.

$$H_{sum} = H_A + H_B + H_C \tag{1}$$

TABLE I
Switching Sequence of VSI corresponding to Hall Effect Signals

Switching Interval	Sequence No.	Position Sensor			Conducting Switch		Phase Currents		
		$H_A = 0$	$H_B = 0$	$H_C = 0$			A	B	C
0°-60°	0	1	0	0	Q1	Q4	$+i_a$	$-i_b$	off
60°-120°	1	1	1	0	Q1	Q6	$+i_a$	off	$-i_c$
120°-180°	2	0	1	0	Q3	Q6	off	$+i_b$	$-i_c$
180°-240°	3	0	1	1	Q3	Q2	$-i_a$	$+i_b$	off
240°-300°	4	0	0	1	Q5	Q2	$-i_a$	off	$+i_c$
300°-360°	5	1	0	1	Q5	Q4	off	$-i_b$	$+i_c$

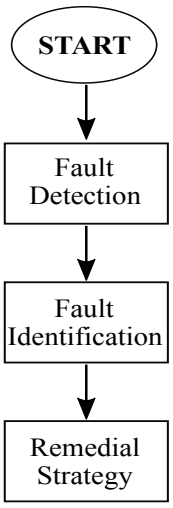


Fig. 3: Proposed fault tolerant control algorithm.

The maximum possible value of H_{sum} is 2, while the minimum possible value is 1. This means that

$$1 \leq H_{sum} \leq 2 \tag{2}$$

If the value of H_{sum} goes beyond this range, it indicates failure in any one of the Hall sensor [11].

B. Fault identification

To identify the faulty sensor, average value of all Hall sensor signals is calculated individually for a few cycles. It can be deduced from output of position sensors given in Table I that in one rotation of motor from 0° to 360°, output of Hall sensor is 1 in three intervals out of total six intervals. Therefore, average value of a healthy signal should be

$$H_{average} = \frac{\text{Sum of Hall sensor outputs during 6 intervals}}{6}$$

$$\frac{3}{6} = 0.5$$

But if the faulty sensor is constantly high, this value will be 1 and if the faulty sensor is constantly low, this value will be 0. Using this algorithm, faulty Hall sensor is identified [11].

C. Remedial Measure: Fictitious Hall Signal Reconstruction

After identification of the faulty Hall sensor, corresponding sensor signal is replaced with a fictitious hall signal which is generated by the microcontroller. The new hall sensor signal is generated based on the phase difference of 120 electrical degrees from the other two Hall sensor signals as shown in Fig. 4. As the motor speed is known and it does not change during fault (speed controller keeps the motor speed constant), the delay time of new hall signal from its preceding hall signal can be determined. Within the microcontroller, this delay is provided using the timer which takes the count value proportional to delay time.

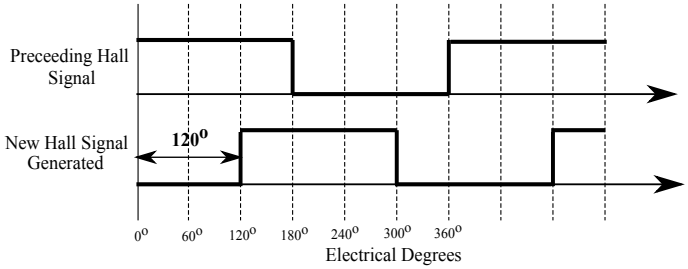


Fig. 4: Generation of new Hall sensor signal.

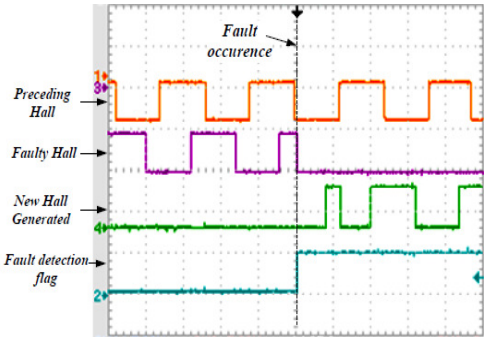


Fig. 5: Detection and replacement of faulty Hall sensor signal.

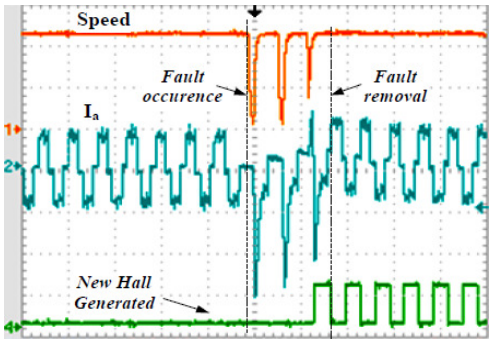


Fig. 6: PMBLDC response during fault diagnosis. Scale : Time = 100 msec/div, Speed = 200 rpm/div, Phase Current = 2 amps/div.

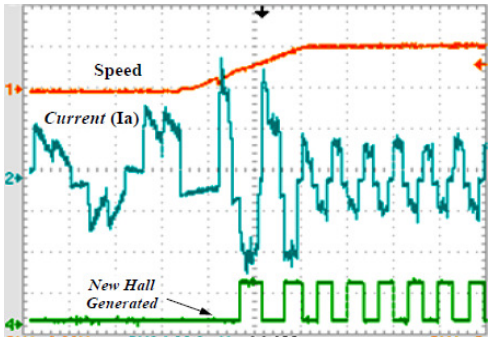
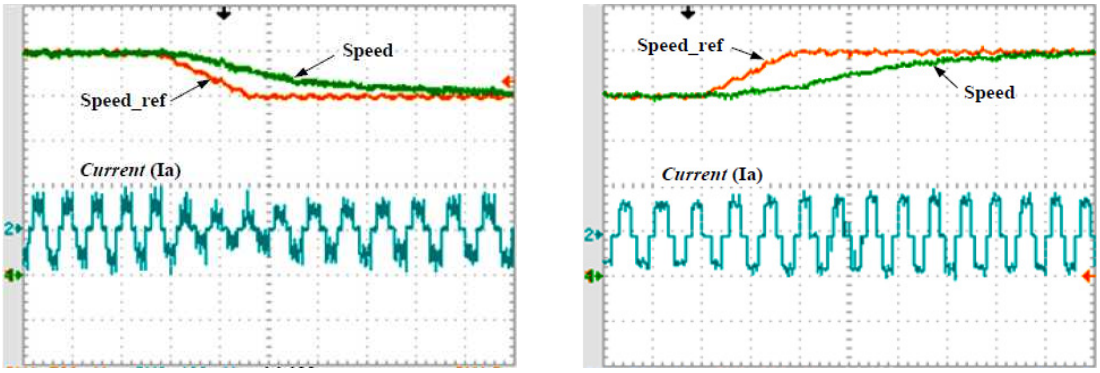
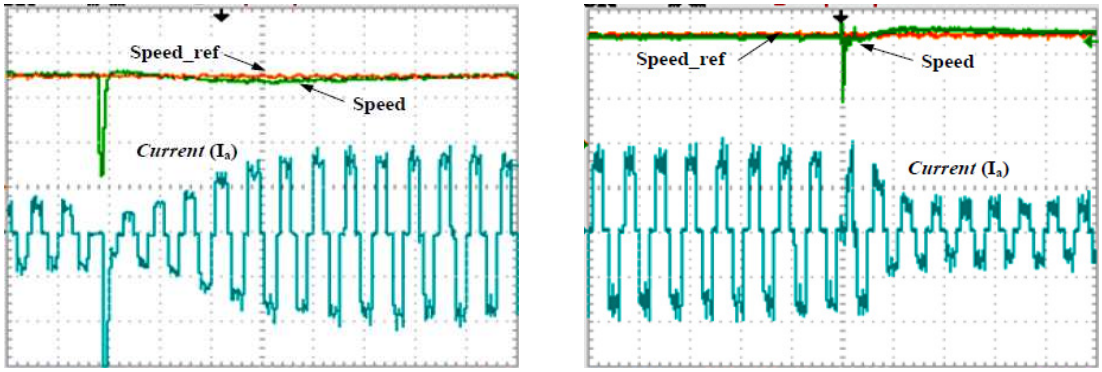


Fig. 7: Starting response of the proposed BLDC drive with one hall sensor fault. Scale : Time = 100 msec/div, Speed = 400 rpm/div, Phase Current = 2 amps/div, Generation of fictitious hall signal = 5 V/div.



(a) (b)

Fig. 8: Transient response during speed change with proposed control during fault. Scale : Speed = 100 rpm/div, Motor phase current : I_a,= 1 A/div, Time : 100 msec/div.



(a) (b)

Fig. 9: Transient response during load change with proposed control during fault. Scale : Speed = 100 rpm/div, Motor phase current : I_a,= 1 A/div, Time : 100 msec/div.

Fig. 5 shows that fault detection algorithm detects the sensor fault almost instantaneously and replaces the faulty sensor with new fictitious hall signal in a very short time. Fig. 6 shows speed and current response with the implemented fault tolerant control system. The machine speed resumes its reference value of 500 rpm in short time and there are no further oscillations in drive speed.

IV. Starting method in the event of failure of one Hall Effect Sensor

After the fault occurrence, PMBLDC motor is unable to self-start with only two hall sensors. Therefore, an algorithm for starting the motor is required. During starting, PMBLDC motor is driven in small steps like a stepper motor for a short duration. Fig. 7 shows the waveforms of current and speed of the motor during starting. In the starting, speed is low and full DC voltage is applied. Motor is observed to take rather large currents at starting but still the peak current value is within safe limits.

V. Transient response of PMBLDC motor with proposed control during fault

Transient response of PMBLDC drive during various conditions is investigated using proposed fault tolerant control. This is done to see how stable is the working of motor using the proposed control. As shown in Fig. 8, motor is responding well to sudden changes in reference speed and there is no overshoot in the current value. Fig. 8(a) and Fig. 8(b) shows the transient behavior for decreasing and increasing reference speed respectively. When the reference speed is changed, the fault control algorithm could modify the count value of timer to change the delay time. Similarly, the response of drive during sudden change in load is shown in Fig. 9. It has been observed that controller is responding well to load changes. Fig. 9(a) and Fig. 9(b) shows the transient behavior for increasing and decreasing load respectively.

VI. Conclusion

Fault tolerant control for compensating the failure of Hall Effect sensors in PMBLDC drive is implemented in a closed loop control scheme. The proposed system is capable to detect and identify the breakdown of sensors and replaces the

faulty sensor signal by an appropriate signal generated by the microcontroller. Effectiveness of the remedial strategy is also established by the correct performance of PMBLDC motor under faulty condition during various transient conditions of load and speed change. The motor is found to be reaching the reference speed with a less settling time and negligible steady state error. The hardware developed is found to be working satisfactorily. The motor is found to be operating smoothly in steady state.

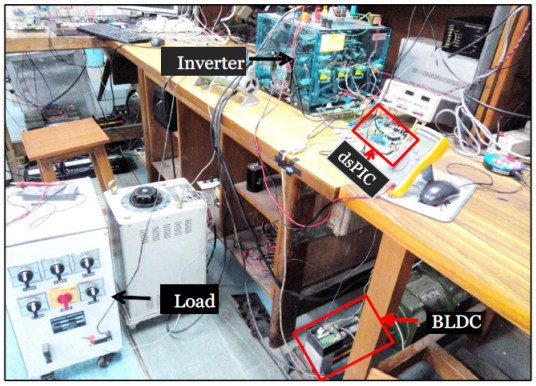
APPENDIX I

Parameters of PMBLDC Motor

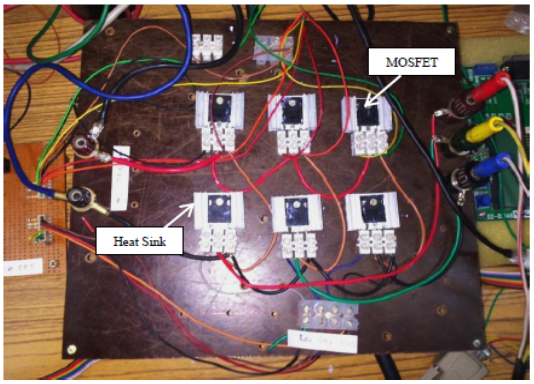
Model	T85.TR.3.2.E.H.15.003.A.16.104
Pole Number	4
Rated Phase to Phase Voltage	540 V
Rated Phase Current	4.3 A
Rated Speed	5200 rpm
Rated Torque	3.2 Nm
Voltage Constant, k_e	78 V/Krpm
Torque Constant, k_t	0.74 Nm/A

APPENDIX II

Experimental Setup



Three Phase MOSFET Inverter



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